

Analyzing the Impact of External Interventions on Water and Energy Demand: Cybersecurity's Role in Sustainable Development

Daryoush Tavangar Rizi

Department of Physics & Energy Engineering
Amirkabir University of Technology
Tehran, Iran
dtavangar@aut.ac.ir

Mohammad Hassan Nazari

Smart Control Systems Department
Niroo Research Institute (NRI)
Tehran, Iran
mhnazari@nri.ac.ir

Ali Maleki

School of Electrical Engineering &
Computer Science
Washington State University
Pullman, Washington
Ali.maleki@wsu.edu

Seyed Hossein Hosseinian

Department of Electrical Engineering
Amirkabir University of Technology
Tehran, Iran
hosseinian@aut.ac.ir

Maryam Fani

Department of Physics & Energy Engineering
Amirkabir University of Technology
Tehran, Iran
mfani@aut.ac.ir

G. B. Gharehpetian

Department of Electrical Engineering
Amirkabir University of Technology
Tehran, Iran
grptian@aut.ac.ir

Abstract—The nexus of water, food, and energy constitutes a fundamental aspect of sustainable development. Furthermore, the demand for water and energy is becoming more pronounced. Moreover, there is a growing importance placed on studying cyber security within the water-energy nexus. This research investigates the interdependence of these two systems, conceptualizing them as a water-energy nexus and effort has been conducted on optimizing the economic efficiency of electricity and water systems in order to minimize operating expenses. Subsequently, external interventions, such as the simultaneous injection of false data on both water and energy demand, were addressed through an optimization process in GAMS software. The results suggest that the value of the cost objective function remains constant when inaccurate data is introduced, which hinders the accurate estimation of the system by the system operator. In this system, both water and power loads are directly impacted by these interventions. In particular, the power and water load of the system have exhibited changes ranging from 1.5% to 17.99% for electric load and 3.87% to 17.1% for water load, in comparison to the previous state.

Keywords—Cyber security, external interventions, optimization, sustainable development, water-energy nexus

I. INTRODUCTION

The prioritization of integrated water-energy management is crucial. Collaborative efforts between decision-makers in both domains are imperative for the management of electric resources and water supply. In [1], a groundbreaking approach to tackling food-water-energy challenges is discussed, shedding light on the relationship between a sustainable environment, water energy, and the development of clean energy infrastructure. An innovative model has been created to establish a connection between water and energy in the integrated energy network, allowing for the effective utilization of renewable energies in the interconnected energy structure, which is presented in [2]. In [3],

a study examined how to optimize the operational plan for merged electricity and water, taking into consideration the interdependencies between them. Considering the change in weather patterns and the increase in crowd size, it is imperative to adopt a planned approach. The electrical and water sectors could benefit from a bottom-up approach that considers the water-energy-emission nexus, as suggested in [4], [5], [6]-[7]. Enhancing smart grid sustainability and resilience by integrating water, energy, and food systems with a practical planning method. It can decrease losses by optimizing, as mentioned in [8], [9]. With the decrease in water resources, using desalination systems in power plants is a good way to make up for it, and there's a specific one mentioned in [10]. Reference [11], there is a system that exemplifies how the control of equipment facilitates the management of electricity and water. Further, the investigation in [12] has centered on the examination of local waste's capability to produce biogas and remediate polluted in water. Nevertheless, the model does not take into consideration the influence of external interventions. False data injection (FDI) is another form of external intervention that could have potentially been employed within the network. In addition, the research conducted in [13] has primarily centered on the analysis of smart grids' functioning during denial of service attacks. In this research, the water-energy nexus networks are at risk from the network attack type investigated as the FDI attack. The accuracy of data [14] has been greatly influenced by this.

The below sections delineate several noteworthy advancements underscored in this research:

- A comprehensive analysis of the cyber security approach includes the evaluation of external interventions and their effects on the demand for electric and water loads at same time.

- The focus of this research is to explore the optimization of electricity and water systems as a means of minimizing operating costs.
- An attempt has been made to use the water for the purpose of as cooler in power plants. On the flip side, electric energy has been used for collecting and purifying water.

The remainder of this study encompasses the achievement of simultaneous electricity and water production in Section I and Section II of the energy grid. Section III contains the analysis and presentation of the results obtained before and after sabotage. Section IV functions as the concluding section of the paper.

II. MODELING SECTION

A. Structure evaluation

The fundamental purpose of this research is:

- The simultaneous distribution of electrical and water demand is being considered within a framework for examining water and energy as central parameters in sustainable development.
- The aim is to conduct experiments on hybrid cyber-attacks, particularly FDI attacks, in order to disrupt the water supply system and energy demand.

The consumption of energy is involved in every stage of the water processing cycle, which includes transportation, water purification for reuse, and collection. Considering the substantial energy requirements associated with water production, it is noteworthy that energy exploitation similarly requires a significant volume of water.

Certain power plants have been compelled to operate at lower capacity in recent years due to water scarcity. Despite their mutual dependence, power and water resources have predominantly remained separate. Despite their mutual dependence, power and water resources have predominantly remained separate. The assessment of water impact is often overlooked during the selection of power plant locations and the determination of energy priorities by politicians.

The research explores the considerations surrounding a power plant that consists of several power generation units and water resources for internal and external consumption. This study delves into the allocation of water resources for a power plant, considering the requirements of both the power plant and consumers outside the complex.

Additionally, the electricity produced is utilized in water purification devices and meets the electrical needs of consumers, thereby aiming to maximize the efficiency of the water-energy system. The example of performing this type of system is illustrated in Fig. 1. The manipulation of measurements in the data, especially data which is essential for network monitoring, has the potential to result in equipment damage caused by FDI attacks. The purpose of this study is to formulate a strategy for external interventions that can impair remote state estimation. In Fig. 2, a small sample of the sabotage performance is shown.

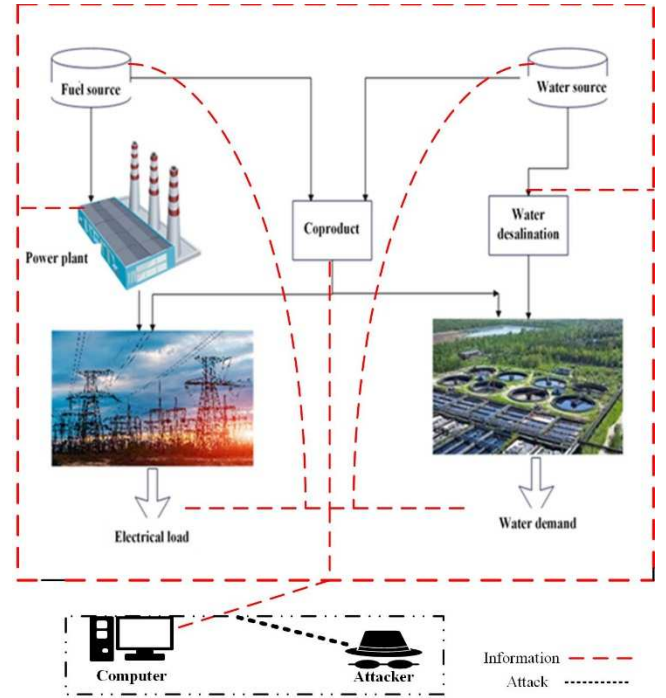


Fig. 1. The system combines water and electricity.

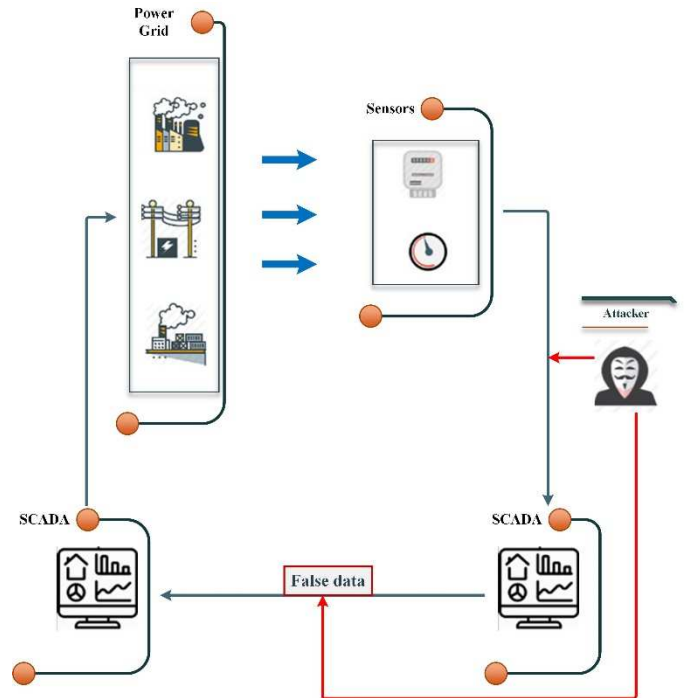


Fig. 2. Attack network information.

B. The objective function in the realm of mathematics.

The overall objective function, as is come in (1), incorporates the total costs of the system, consisting of three components: coproduction costs, power generation plant costs, and water production costs [15].

$$\min_{set} (\text{Total Cost}) = CCo + CPGP + CWP \quad (1)$$

The cost of coproduction, power generation plant, and water production are represented by CCo , $CPGP$, and CWP , respectively. The decision variables A, B, and C correspond to the power generated by the power plant at time E. Costs in (2)-(3)-(4) are planned separately within the following relationships.

$$CPGP = a_g x_{g,t}^2 + b_g x_{g,t} + c_g z_{g,t} \quad (2)$$

$$CCo = \alpha_c x_{c,t}^2 + \beta_c x_{c,t} + \gamma_c x_{c,t}^2 + \zeta_c x_{c,t} + \xi_c x_{c,t} + \lambda_c z_{c,t} \quad (3)$$

$$CWP = a_w x_{w,t}^2 + b_w x_{w,t} + c_w z_{w,t} \quad (4)$$

C. Constraints imposed on the objective function

This section outlines the applicable constraints. Initially, the technical limitations regarding the upper and lower bounds of power and water production units are provided in (5)-(6)-(7):

$$x_g^{\min} z_{g,t} \leq x_{g,t} \leq x_g^{\max} z_{g,t} \quad (5)$$

$$x_c^{\min} z_{c,t} \leq x_{c,t} \leq x_c^{\max} z_{c,t} \quad (6)$$

$$x_w^{\min} z_{w,t} \leq x_{w,t} \leq x_w^{\max} z_{w,t} \quad (7)$$

The provision of electricity necessitates the fulfillment of the power balance condition, expressed in (8). It is necessary for the hourly production to correspond to the consumption of each load. Furthermore, the equilibrium requirement for hourly water demand is specified in equation (9):

$$p_t^{elec-load} = \sum_g x_{g,t} + \sum_c x_{c,t} \quad (8)$$

$$p_t^{water-demand} = \sum_w x_{w,t} + \sum_c x_{c,t} \quad (9)$$

The second scenario involves constraints that revolve around cyber interventions, namely the deliberate injection of false data into the electrical and water loads of the system in (10)-(11)-(12).

$$\sum_{t=1}^{24} \Delta D_t^{elec-load} = 0 \quad (10)$$

$$-v \times p_t^{elec-load} \leq \Delta D_t^{elec-load} \leq v \times p_t^{elec-load} \quad (11)$$

$$p_t^{elec-load} + \Delta D_t^{elec-load} = \sum_g x_{g,t} + \sum_c x_{c,t} \quad (12)$$

$\Delta D_t^{elec-load}$ represents incorrect injection data, while $p_t^{elec-load}$ represents the demand value of the electric load. Likewise, interventions are defined and implemented on water demand, as indicated by equations (13)-(14)-(15).

$$\sum_{t=1}^{24} \Delta D_t^{water-demand} = 0 \quad (13)$$

$$-v \times p_t^{water-demand} \leq \Delta D_t^{water-demand} \leq v \times p_t^{water-demand} \quad (14)$$

$$p_t^{water-demand} + \Delta D_t^{water-demand} = \sum_w x_{w,t} + \sum_c x_{c,t} \quad (15)$$

III. SIMULATION RESULTS

The solution is recognized as utilizing a two-step approach. The solution is recognized as utilizing a two-step approach. The initial stage represents the power-water connection under normal network conditions, whereas the subsequent stage incorporates an erroneous data injection model into the optimization problem for electric and water loads. A thorough analysis has been performed to investigate the cost and electrical load alterations related to the scenarios before and after the external interventions. The data regarding the electrical load is depicted in Fig. 3. Also, the water demand of the system is visualized in Fig. 4.

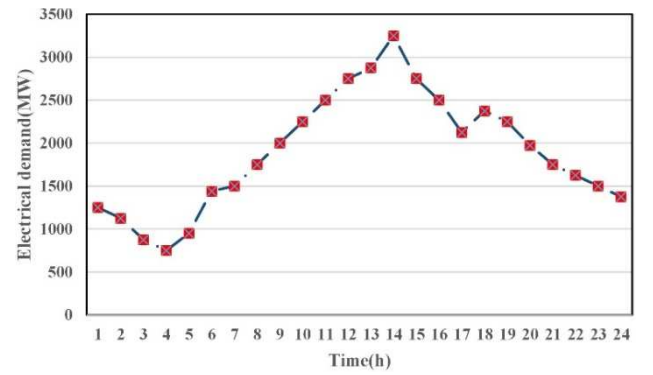


Fig. 3. The demand for electricity consumption..

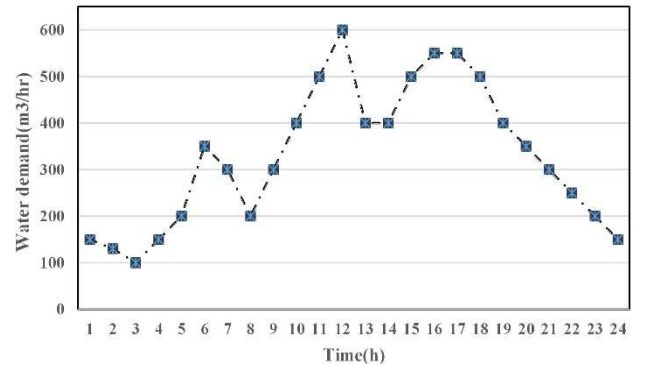


Fig. 4. The demand for water consumption.

TABLE I. COSTS OF TWO DIFFERENT SITUATIONS WITH THE AIM OF SABOTAGES

Costs	Cost before and after a attack		
	<i>the system without FDI</i>	<i>the system with FDI</i>	<i>difference cost</i>
Operating cost (\$)	28359.671	28359.671	0

TABLE II. INCORECT DATA INJECTION FOR ELECTRICITY LOAD

Time	false injected data	Time	false injected data
1	0	13	-373.321
2	193.216	14	-499.084
3	-26.558	15	-73.655
4	128.810	16	263.678
5	98.709	17	288.678
6	-19.198	18	-48.655
7	68.537	19	49.698
8	-100	20	-54.198
9	-31.464	21	-181.464
10	100.916	22	-116.378
11	176.345	23	-181.949
12	101.011	24	236.324
Total in 24 hours		0	

TABLE III. CHANGING ELECTRICITY LOAD

Time	Manipulated load data	Time	Manipulated load data
1	1250	13	2501.679
2	1318.216	14	2750.916
3	848.442	15	2676.345
4	878.81	16	2763.678
5	1048.709	17	2413.678
6	1420.802	18	2326.345
7	1568.537	19	2299.698
8	1650	20	1920.802
9	1968.536	21	1568.536
10	2350.916	22	1508.622
11	2676.345	23	1318.051
12	2851.011	24	1612.324

According to Table I, the total system costs in both scenarios have reached a value of 28,359.671 dollars. The equalized cost is the reason operator can't detect it. The incorrect representation of injected power in the information pertaining to network electric loads is evident from Table II.

Furthermore, it is shown that the total injection data for the day is equivalent to zero. Table III is illustrated the extent of inaccurate data within the varying electricity load. in Fig. 9 is presented a combination of healthy and unhealthy data.

Similarly, the assumption has been made that this type of external intervention has occurred in the system regarding water demand. According to the data is presented in Table IV, there has been of incorrect data injection during the scheduled time.

Furthermore, Table V has provided an estimation of the quantity of water demand that is incorrectly assigned.

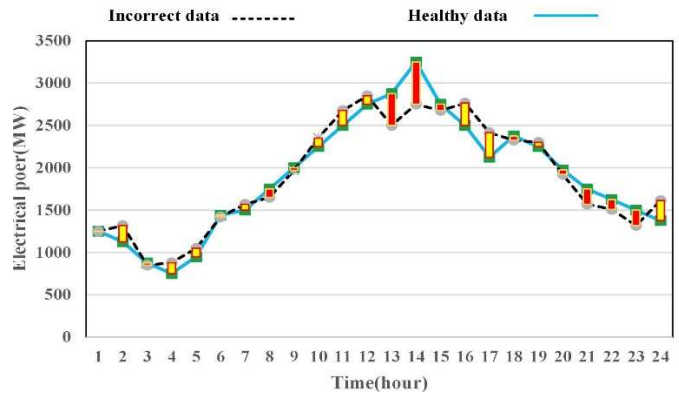


Fig. 5. Comparative of electrical demand in two step.

TABLE IV. INCORECT DATA INJECTION FOR WATER DEMAND

Time	false injected data	Time	false injected data
1	25.762	13	-68.699
2	22.327	14	68.699
3	17.175	15	-28.807
4	25.762	16	-84.503
5	34.349	17	-10.057
6	-60.111	18	-85.874
7	51.524	19	-50
8	34.349	20	29.391
9	51.524	21	46.193
10	15.497	22	42.937
11	-34.503	23	34.349
12	-103.048	24	25.762
Total in 24 hours		0	

TABLE V. CHANGING WATER DEMAND

Time	Manipulated load data	Time	Manipulated load data
1	124.238	13	468.699
2	107.673	14	331.301
3	82.825	15	528.807
4	124.238	16	634.503
5	165.651	17	560.057
6	410.111	18	585.874
7	248.476	19	450
8	165.651	20	320.609
9	248.476	21	253.807
10	384.503	22	207.063
11	534.503	23	165.651
12	703.048	24	124.238

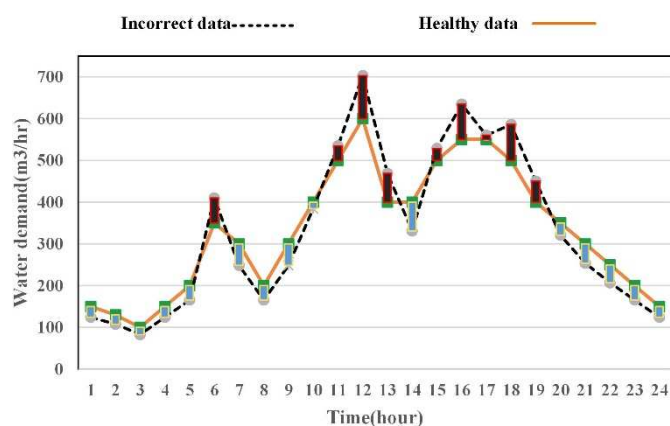


Fig. 6. Comparative of water demand in two step.

The data is shown in Fig. 10 includes both correct and incorrect information.

IV. CONCLUSION

The integration of surplus water sources has played a significant role in enhancing energy generation and conversion within intelligent energy networks, particularly in power plants. The focus of this research has been to explore the interdependence of these systems, treating them as a water-energy nexus. Its main aim has been to maximize economic efficiency while minimizing operating expenses in electricity and water systems. The results are demonstrated that the meters indicate a higher water volume during peak hours, exceeding the system's necessary values. Specifically, the system has encountered variations in power and water load, ranging from 1.5% to 17.99% for electric load and 3.87% to 17.1% for water load, in comparison to the previous condition. Furthermore, the identification of the attacks has proven challenging due to their simultaneous occurrence and combination.

REFERENCES

- [1] F. Meng, Q. Yuan, R. A. Bellezoni, J. A. Puppim de Oliveira, S. Cristiano, A. M. Shah, G. Liu, Z. Yang, Karen C. Seto, "Quantification of the food-water-energy nexus in urban green and blue infrastructure: A synthesis of the literature," *Resources, Conservation and Recycling*, Vol. 188, pp. 106658, 2023, doi: 10.1016/j.resconrec.2022.106658
- [2] M. Daneshvar, B. Mohammadi-ivatloo and A. Anvari-Moghaddam, "Risk-Oriented Operational Model for Fully Renewable Cooperative Prosumers in a Modern Water-Energy Nexus Structure," in *IEEE Transactions on Sustainable Energy*, vol. 14, no. 4, pp. 2006-2019, Oct. 2023, doi: 10.1109/TSTE.2023.3287768.
- [3] A. S. Zamzam, E. Dall'Anese, C. Zhao, J. A. Taylor and N. D. Sidiropoulos, "Optimal Water-Power Flow-Problem: Formulation and Distributed Optimal Solution," in *IEEE Transactions on Control*

- of Network Systems, vol. 6, no. 1, pp. 37-47, March 2019, doi: 10.1109/TCNS.2018.2792699.
- [4] M. H. Nazari, S. H. Hosseini, E. A. Farsani, "An incentive-based policy on minimization of GHG emissions and loss using adaptive group search multi-objective optimization algorithm," *Scientia Iranica, Vol. 29, Issue 1, Pages 230-246*, 2022. DOI: 10.24200/sci.2019.53554.3326. https://scientiainiranica.sharif.edu/article_21546.html
- [5] H. Khajepour, N. Norouzi, and M. Fani, "An Exergetic Model for the Ambient Air Temperature Impacts on the Combined Power Plants and its Management Using the Genetic Algorithm," vol. 29, no. 1, Feb. 2021, doi: 10.1142/S2010132521500085.
- [6] F. Afsari Mamaghani, B. Omidvar, A. Avami, and G. Nabi-Bidhendi, "An optimal integrated power and water supply planning model considering Water-Energy-Emission nexus," *Energy Convers Manag.*, vol. 277, p. 116595, Feb. 2023, doi: 10.1016/J.ENCONMAN.2022.116595.
- [7] F. Afsari Mamaghani, B. Omidvar, A. Avami, Gh. Nabi-Bidhendi, "An optimal integrated power and water supply planning model considering Water-Energy-Emission nexus," *Energy Conversion and Management*, 2023. Available: <https://www.sciencedirect.com/science/article/pii/S0196890422013735>
- [8] Shang-Keng Liu, Zih-Ee Lin, Pei-Te Chiueh, "Improving urban sustainability and resilience with the optimal arrangement of water-energy-food related practices," *Science of The Total Environment*, Vol. 812, pp. 152559, 2023. Doi: 10.1016/j.scitotenv.2021.152559.
- [9] M. H. Nazari, S. H. Hosseini, and E. Azad-Farsani, "Shapley Value-Based Techno-Economic Framework for Harmonic and Loss Mitigation," *IEEE Access*, vol. 7, pp. 119576-119592, Aug. 2019, doi: 10.1109/access.2019.2936859.
- [10] Aref Shokri, Mahdi Sanavi Fard, "Water-energy nexus: Cutting edge water desalination technologies and hybridized renewable-assisted systems; challenges and future roadmaps," *Sustainable Energy Technologies and Assessments*, Vol. 57, pp. 103173, doi: 10.1016/j.seta.2023.103173.
- [11] Borhan Soleimani, Dariush Keihan Asl, Javad Estakhr, Ali Reza Seifi, "Integrated optimization of multi-carrier energy systems: Water-energy nexus case," *Energy*, Vol. 257, pp. 124764, 2022, doi: 10.1016/j.energy.2022.124764.
- [12] Sepideh Abedi, Azadeh Nozarpour, Omid Tavakoli, "Evaluation of biogas production rate and leachate treatment in landfill through a water-energy nexus framework for integrated waste management," *Energy Nexus*, Vol. 11, pp. 100218, 2023, doi: 10.1016/j.nexus.2023.100218.
- [13] M. Jamali, H. R. Baghaee, M. S. Sadabadi, G. B. Gharehpetian, and A. Anvari-Moghaddam, "Distributed Cooperative Event-Triggered Control of Cyber-Physical AC Microgrids Subject to Denial-of-Service Attacks," *IEEE Trans Smart Grid*, Nov. 2023, doi: 10.1109/TSG.2023.3259545.
- [14] Xueping Li, Yaokun Wang, Zhigang Lu, "Graph-based detection for false data injection attacks in power grid," *Energy*, vol. 263, Part C, pp. 125865, 2023, doi: 10.1016/j.energy.2022.125865.
- [15] Apoorva Santhosh, Amro M. Farid, Kamal Youcef-Toumi, "Real-time economic dispatch for the supply side of the energy-water nexus," *Applied Energy*, vol. 122, pp. 42-52, 2014, doi: 10.1016/j.apenergy.2014.01.062.